



KALLAM HARANADHAREDDY INSTITUTE OF TECHNOLOGY
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CODING QUESTIONS FOR ACCENTURE DRIVE

Test	Coding
No. of questions	2
Difficulty level	high
Salary	4.5-6.5 lacs Per Annum
Time limit	45 minutes

Accenture Coding Questions

This section is there to test out your coding and problem-solving techniques. There will 2 problem statements, for which you have to provide a code solution in any of the below-mentioned languages

- C
- C++
- Java
- Python
- Dot Net

Some of the topics that you should be knowing are:

- Data Structures
- Greedy and Dynamic Programming
- Hashing
- Maps

1. Reverse a String

- Write a function to reverse a string without using built-in functions.



2. Find the Missing Number

- Given an array of $n-1$ integers in the range 1 to n , find the missing number.

3. Find the Largest Element in an Array

- Write a function to find the largest element in an array.

4. Check for Palindrome

- Determine if a given string is a palindrome.

5. Merge Two Sorted Arrays

- Merge two sorted arrays into one sorted array.

6. Fibonacci Series

- Write a function to generate the Fibonacci series up to n terms.

7. Find the Intersection of Two Arrays

- Given two arrays, find the intersection of the two arrays.

8. Count Occurrences of a Character

- Count the number of occurrences of a specific character in a string.

9. Find the Second Largest Element

- Write a function to find the second largest element in an array.

10. Check for Balanced Parentheses

Determine if a string containing parentheses is balanced.

11. Find All Prime Numbers in a Range

Write a function to find all prime numbers in a given range.

12. Sort an Array Using Bubble Sort

Implement bubble sort to sort an array.

13. Find the GCD of Two Numbers

Calculate the Greatest Common Divisor (GCD) of two numbers.

14. Find the Sum of Digits

Write a function to find the sum of digits of a given number.

15. Find the Longest Common Prefix

Write a function to find the longest common prefix string amongst an array of strings.

16. Find the Duplicate in an Array



Given an array of n integers where each integer is between 1 and $n-1$, find the duplicate.

17. Rotate an Array

Rotate an array to the right by k steps.

18. Find the Kth Largest Element

Write a function to find the k^{th} largest element in an array.

19. Check if Two Strings are Anagrams

Determine if two strings are anagrams of each other.

20. Implement a Linked List

Implement a singly linked list with basic operations like insertion, deletion, and traversal.

21. Detect a Cycle in a Linked List

Write a function to detect if a cycle exists in a linked list.

22. Reverse a Linked List

Reverse a singly linked list.

23. Find the Intersection Point in Two Linked Lists

Determine the intersection point of two linked lists.

24. Add Two Numbers Represented by Linked Lists

Add two numbers represented by linked lists.

25. Binary Search

Implement binary search to find a target value in a sorted array.

26. Find the Height of a Binary Tree

Write a function to find the height of a binary tree.

27. Inorder Traversal of a Binary Tree

Perform an inorder traversal of a binary tree.

28. Check if a Binary Tree is Balanced

Determine if a binary tree is height-balanced.

29. Convert a Binary Tree to a Doubly Linked List

Convert a binary tree into a doubly linked list in-place.

30. Generate Pascal's Triangle

Write a function to generate the first n rows of Pascal's Triangle.



31. Find All Subsets of a Set

Given a set of integers, find all possible subsets.

32. Find the Longest Increasing Subsequence

Write a function to find the length of the longest increasing subsequence.

33. Find the Median of Two Sorted Arrays

Find the median of two sorted arrays of the same size.

34. Implement a Stack Using Arrays

Implement a stack data structure using arrays with push, pop, and peek operations.

35. Implement a Queue Using Stacks

Implement a queue using two stacks.

36. Find the Shortest Path in a Graph

Implement Dijkstra's algorithm to find the shortest path in a graph.

37. Solve the N-Queens Problem

Write a function to solve the N-Queens problem.

38. Find the Longest Palindromic Substring

Write a function to find the longest palindromic substring in a given string.

39. Count the Number of Islands

Given a 2D grid of 1s (land) and 0s (water), count the number of islands.

40. Find All Permutations of a String

Write a function to find all permutations of a given string.

41. Implement Quick Sort

Write a function to sort an array using Quick Sort.

42. Find the Most Frequent Element

Find the most frequent element in an array.

43. Determine if a String is a Valid Number

Check if a given string represents a valid number (integer or floating-point).

44. Find the Sum of Elements in a Binary Tree

Calculate the sum of all elements in a binary tree.

45. Find All Anagrams in a String



Given a string, find all substrings that are anagrams of a given pattern.

46. Check if a Linked List is a Palindrome

Determine if a linked list is a palindrome.

47. Rotate a Matrix

Rotate a given $n \times n$ matrix by 90 degrees clockwise.

48. Implement a Trie Data Structure

Implement a trie with insert and search operations.

49. Find the Largest Rectangle in Histogram

Given a histogram represented by an array of heights, find the area of the largest rectangle that can be formed.

50. Compute the Power of a Number

Implement a function to compute x^n raised to the power of n using recursion or iteration.

1. Find the Unique Element in an Array

- Given an array where every element occurs twice except one, find the unique element.

2. Calculate the Power of a Number

- Write a function to compute x raised to the power n without using built-in power functions.

3. Find the Median of a List

- Given a list of integers, find the median.

4. Implement Depth-First Search (DFS)

- Implement DFS for a graph using recursion.

5. Implement Breadth-First Search (BFS)

- Implement BFS for a graph using a queue.

6. Check for a Subarray with Zero Sum

- Determine if there is a subarray within a given array that sums up to zero.

7. Find the Longest Common Subsequence

- Write a function to find the length of the longest common subsequence between two strings.

8. Find the Smallest Positive Missing Number

- Given an unsorted array of integers, find the smallest positive missing number.



9. Find the kth Smallest Element in an Array

- Implement a function to find the kth smallest element in an array.

10. Find All Unique Subsets

Given a set of integers, find all unique subsets of the set.

11. Find the Minimum Depth of a Binary Tree

Write a function to find the minimum depth of a binary tree.

12. Find the Maximum Depth of a Binary Tree

Determine the maximum depth of a binary tree.

13. Check if a Binary Tree is Symmetric

Write a function to check if a binary tree is symmetric around its center.

14. Implement a Priority Queue

Implement a priority queue using a heap data structure.

15. Count the Number of Paths in a Grid

Find the number of ways to reach the bottom-right corner of a grid from the top-left corner.

16. Find All Combinations of a Set

Given a set of integers, find all combinations of elements that sum up to a target value.

17. Find the Longest Substring Without Repeating Characters

Write a function to find the length of the longest substring without repeating characters.

18. Calculate the nth Fibonacci Number Using Dynamic Programming

Compute the n^{th} Fibonacci number using dynamic programming.

19. Determine if a Binary Tree is a Binary Search Tree (BST)

Check if a binary tree is a valid binary search tree.

20. Find the Shortest Path in a Weighted Graph

Implement Dijkstra's algorithm to find the shortest path in a weighted graph.

21. Implement Merge Sort

Write a function to sort an array using Merge Sort.

22. Check for Valid Sudoku

Validate if a given Sudoku board configuration is valid.

23. Find the Closest Pair of Points



Given a set of points, find the pair of points with the smallest distance between them.

24. Find the Longest Increasing Subsequence in a Matrix

Given a matrix, find the longest increasing subsequence.

25. Find the Peak Element in an Array

Find an element in an array that is greater than its neighbors.

26. Find All Permutations of a String with Duplicates

Given a string with duplicate characters, find all unique permutations.

27. Check if a Graph is Bipartite

Determine if a given graph is bipartite.

28. Calculate the Number of Ways to Make Change

Given an array of coin denominations, find the number of ways to make change for a given amount.

29. Find the Largest Rectangle in a Binary Matrix

Given a binary matrix, find the area of the largest rectangle of 1s.

30. Implement a Circular Queue

Implement a circular queue using arrays.

31. Find the Maximum Product Subarray

Given an array of integers, find the contiguous subarray with the maximum product.

32. Find the Median of Two Sorted Arrays

Write a function to find the median of two sorted arrays of different sizes.

33. Find All Substrings of a String

Generate all possible substrings of a given string.

34. Find the Most Frequent Substring

Given a string, find the most frequent substring of a given length.

35. Count the Number of Palindromic Substrings

Count the number of distinct palindromic substrings in a given string.

36. Find the Longest Repeating Substring

Given a string, find the longest substring that appears more than once.

37. Find the First Non-Repeating Character



Given a string, find the first non-repeating character.

38. Implement a Hash Map

Implement a basic hash map with operations like insert, delete, and search.

39. Solve the N-Queens Problem Using Backtracking

Write a function to solve the N-Queens problem using backtracking.

40. Implement Depth-First Search (DFS) for a Maze

Use DFS to solve a maze problem and find a path from start to end.

41. Find the Largest Sum of Contiguous Subarray (Kadane's Algorithm)

Write a function to find the maximum sum of a contiguous subarray using Kadane's Algorithm.

42. Determine if a String is a Valid IP Address

Check if a given string is a valid IPv4 or IPv6 address.

43. Find the Minimum Window Substring

Given a string and a pattern, find the minimum window substring that contains all characters of the pattern.

44. Generate Parentheses

Generate all valid combinations of `n` pairs of parentheses.

45. Find the Shortest Subarray with Sum Greater than a Given Value

Write a function to find the shortest subarray whose sum is greater than a given value.

46. Find the Maximum Sum of Non-Adjacent Elements

Given an array of integers, find the maximum sum of non-adjacent elements.

47. Calculate the Length of Longest Palindromic Subsequence

Find the length of the longest palindromic subsequence in a string.

48. Find the Lowest Common Ancestor in a Binary Tree

Write a function to find the lowest common ancestor of two nodes in a binary tree.

49. Implement a Simple File Compression Algorithm

Implement a basic file compression algorithm such as Run-Length Encoding (RLE).

50. Check if a String is a Valid Anagram

Given two strings, check if they are anagrams of each other by comparing character frequencies

Previous Accenture Drive Coding Questions:



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Coding Question 1

You are required to implement the following function:

def LargeSmallSum(arr):

The function accepts an integer array **arr** of size **length** as its argument. You are required to return the sum of the second largest element from the even positions and the second smallest element from the odd positions of the given **arr**.

Assumptions:

- All array elements are unique.
- Treat the 0th position as even.

Notes:

- Return 0 if the array is empty.
- Return 0 if the array length is 3 or less than 3.

Example:

Input:

- **arr: 4 1 8 7 2 9**

Output:

- **9**

Explanation:

- Second largest among even position elements (4, 8, 2) is 4.
- Second smallest among odd position elements (1, 7, 9) is 7.
- Thus, the output is $4 + 7 = 11$.

Sample Input:

- **arr: 9 3 5 8 6 2 7**

Sample Output:

- **10**

Coding Question 2

You are required to implement the following function:

def LargeSmallSum(arr):



The function accepts an integer array `arr` of size `length` as its argument. You are required to return the sum of the second largest element from the even positions and the second smallest element from the odd positions of the given `arr`.

Assumptions:

- All array elements are unique.
- Treat the 0th position as even.

Notes:

- Return 0 if the array is empty.
- Return 0 if the array length is 3 or less than 3.

Example:

Input:

- `arr: 10 2 6 4 9 7`

Output:

- `12`

Explanation:

- Second largest among even position elements (10, 6, 9) is 9.
- Second smallest among odd position elements (2, 4, 7) is 4.
- Thus, the output is $9 + 4 = 13$.

Sample Input:

- `arr: 11 3 14 6 8 1 5`

Sample Output:

- `15`

Coding Question 3

The function `def differenceofSum(n, m)` accepts two integers `n` and `m` as arguments. Find the sum of all numbers in the range from 1 to `m` (both inclusive) that are not divisible by `n`. Return the difference between the sum of integers not divisible by `n` and the sum of numbers divisible by `n`.

Assumption:

- $n > 0$ and $m > 0$
- Sum lies within the integral range



Example:

- **Input:**
n: 5
m: 25
 - **Output:**
130
 - **Explanation:**
Sum of numbers divisible by 5: $5 + 10 + 15 + 20 + 25 = 75$
Sum of numbers not divisible by 5: $1 + 2 + 3 + 4 + 6 + 7 + 8 + 9 + 11 + 12 + 13 + 14 + 16 + 17 + 18 + 19 + 21 + 22 + 23 + 24 = 205$
Difference: $205 - 75 = 130$
-

Coding Question 4

The function `def differenceofSum(n, m)` accepts two integers `n` and `m` as arguments. Find the sum of all numbers in the range from 1 to `m` (both inclusive) that are not divisible by `n`. Return the difference between the sum of integers not divisible by `n` and the sum of numbers divisible by `n`.

Assumption:

- $n > 0$ and $m > 0$
- Sum lies within the integral range

Example:

- **Input:**
n: 7
m: 35
 - **Output:**
350
 - **Explanation:**
Sum of numbers divisible by 7: $7 + 14 + 21 + 28 + 35 = 105$
Sum of numbers not divisible by 7: $1 + 2 + 3 + 4 + 5 + 6 + 8 + 9 + 10 + 11 + 12 + 13 + 15 + 16 + 17 + 18 + 19 + 20 + 22 + 23 + 24 + 25 + 26 + 27 + 29 + 30 + 31 + 32 + 33 + 34 = 455$
Difference: $455 - 105 = 350$
-

Coding Question 5

The function `def differenceofSum(n, m)` accepts two integers `n` and `m` as arguments. Find the sum of all numbers in the range from 1 to `m` (both inclusive) that are not divisible by `n`. Return the difference between the sum of integers not divisible by `n` and the sum of numbers divisible by `n`.



Assumption:

- $n > 0$ and $m > 0$
- Sum lies within the integral range

Example:

- **Input:**
n: 6
m: 18
- **Output:**
72
- **Explanation:**
Sum of numbers divisible by 6: $6 + 12 + 18 = 36$
Sum of numbers not divisible by 6: $1 + 2 + 3 + 4 + 5 + 7 + 8 + 9 + 10 + 11 + 13 + 14 + 15 + 16 + 17 = 108$
Difference: $108 - 36 = 72$

Coding Question 6

The function `def differenceofSum(n, m)` accepts two integers `n` and `m` as arguments. Find the sum of all numbers in the range from 1 to `m` (both inclusive) that are not divisible by `n`. Return the difference between the sum of integers not divisible by `n` and the sum of numbers divisible by `n`.

Assumption:

- $n > 0$ and $m > 0$
- Sum lies within the integral range

Example:

- **Input:**
n: 2
m: 10
- **Output:**
25
- **Explanation:**
Sum of numbers divisible by 2: $2 + 4 + 6 + 8 + 10 = 30$
Sum of numbers not divisible by 2: $1 + 3 + 5 + 7 + 9 = 25$
Difference: $25 - 30 = -5$

Coding Question 7



The function `def differenceofSum(n, m)` accepts two integers `n` and `m` as arguments. Find the sum of all numbers in the range from 1 to `m` (both inclusive) that are not divisible by `n`. Return the difference between the sum of integers not divisible by `n` and the sum of numbers divisible by `n`.

Assumption:

- $n > 0$ and $m > 0$
- Sum lies within the integral range

Example:

- Input:
n: 9
m: 30
- Output:
238
- Explanation:
Sum of numbers divisible by 9: $9 + 18 + 27 = 54$
Sum of numbers not divisible by 9: $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 10 + 11 + 12 + 13 + 14 + 15 + 16 + 17 + 19 + 20 + 21 + 22 + 23 + 24 + 25 + 26 + 28 + 29 + 30 = 292$
Difference: $292 - 54 = 238$

Coding Question 8

The function `def differenceofSum(n, m)` accepts two integers `n`, `m` as arguments Find the sum of all numbers in range from 1 to `m`(both inclusive) that are not divisible by `n`. Return difference between sum of integers not divisible by `n` with sum of numbers divisible by `n`.

Assumption: $n > 0$ and $m > 0$ Sum lies between integral range

Example Input n:4 m:20

Output 90

Explanation Sum of numbers divisible by 4 are $4 + 8 + 12 + 16 + 20 = 60$ Sum of numbers not divisible by 4 are $1 + 2 + 3 + 5 + 6 + 7 + 9 + 10 + 11 + 13 + 14 + 15 + 17 + 18 + 19 = 150$ Difference $150 - 60 = 90$

Sample Input n:3 m:10

Sample Output 19

Coding Question 9

Implement the following function:



def ProductSmallestPair(sum, arr):

The function accepts an integer sum and an integer array arr of size n. Implement the function to find the pair (arr[j], arr[k]) where $j \neq k$, such that arr[j] and arr[k] are the least two elements of the array, and $(arr[j] + arr[k] \leq sum)$. Return the product of the elements of this pair.

Notes:

- Return -1 if the array is empty or if $n < 2$.
- Return 0 if no such pairs are found.
- All computed values lie within the integer range.

Example:

Input:

- sum: 10
- arr: 4 2 7 1 5

Output:

- 2

Explanation:

- Pair of the least two elements is (1, 2).
- $1 + 2 = 3 \leq 10$, so the product is $1 * 2 = 2$. Thus, the output is 2.

Sample Input:

- sum: 3
- arr: 5 9 8 12

Sample Output:

- 0

Coding Question 10

Implement the following function:

def ProductSmallestPair(sum, arr):

The function accepts an integer sum and an integer array arr of size n. Implement the function to find the pair (arr[j], arr[k]) where $j \neq k$, such that arr[j] and arr[k] are the least two elements of the array, and $(arr[j] + arr[k] \leq sum)$. Return the product of the elements of this pair.

Notes:



- Return -1 if the array is empty or if $n < 2$.
- Return 0 if no such pairs are found.
- All computed values lie within the integer range.

Example:

Input:

- sum: 15
- arr: 10 6 4 8 2

Output:

- 8

Explanation:

- Pair of the least two elements is (2, 4).
- $2 + 4 = 6 \leq 15$, so the product is $2 * 4 = 8$. Thus, the output is 8.

Sample Input:

- sum: 5
- arr: 20 18 15 10

Sample Output:

- 0

Coding Question 11

Implement the following function:

```
def ProductSmallestPair(sum, arr):
```

The function accepts an integer sum and an integer array arr of size n. Implement the function to find the pair (arr[j], arr[k]) where $j \neq k$, such that arr[j] and arr[k] are the least two elements of the array, and (arr[j] + arr[k] $\leq$ sum). Return the product of the elements of this pair.

Notes:

- Return -1 if the array is empty or if $n < 2$.
- Return 0 if no such pairs are found.
- All computed values lie within the integer range.

Example:



Input:

- sum: 20
- arr: 12 8 6 3 11 5

Output:

- 15

Explanation:

- Pair of the least two elements is (3, 5).
- $3 + 5 = 8 \leq 20$, so the product is $3 * 5 = 15$. Thus, the output is 15.

Sample Input:

- sum: 6
- arr: 7 6 4

Sample Output:

- 0

Coding Question: 12

Binary to Decimal Conversion

Problem Statement:

Binary notation is a system for representing numbers using only two symbols: 0 and 1. In this problem, you need to convert a binary number to its decimal equivalent.

Implement the following function:

```
int BinaryToDecimal(const char* binaryStr);
```

The function accepts a string `binaryStr`, representing a binary number. Implement the function to calculate the decimal equivalent of the binary number and return the same as an integer.

Steps:

1. Start with the least significant digit (rightmost digit) of the binary string.
2. For each digit, multiply it by 2 raised to the power of its position index (starting from 0).
3. Sum all the results to get the decimal equivalent.

Assumption:

The input string contains only '0' and '1' characters.



Example

Input:

- `binaryStr: "1101"`

Output:

- `13`

Explanation:

- $1 * 2^3 + 1 * 2^2 + 0 * 2^1 + 1 * 2^0 = 8 + 4 + 0 + 1 = 13$
-

Coding Question: 13

Octal to Decimal Conversion

Problem Statement:

Octal notation is a base-8 system, using digits 0 through 7. In this problem, you need to convert an octal number to its decimal equivalent.

Implement the following function:

```
int OctalToDecimal(const char* octalStr);
```

The function accepts a string `octalStr`, representing an octal number. Implement the function to calculate the decimal equivalent of the octal number and return the same as an integer.

Steps:

1. Start with the least significant digit (rightmost digit) of the octal string.
2. For each digit, multiply it by 8 raised to the power of its position index (starting from 0).
3. Sum all the results to get the decimal equivalent.

Assumption:

The input string contains only '0' to '7' characters.

Example

Input:

- `octalStr: "157"`

Output:

- `111`

Explanation:

- $1 * 8^2 + 5 * 8^1 + 7 * 8^0 = 64 + 40 + 7 = 111$

Coding Question: 14

N-base Addition

Problem Statement:

You are given two numbers in n-base notation and need to find their sum, also in n-base notation.

Implement the following function:

```
char* AddNBase(int n, const char* num1, const char* num2);
```

The function accepts an integer n and two strings num1 and num2, representing two numbers in n-base notation. Implement the function to calculate the sum of these two numbers and return the result as a string in n-base notation.

Steps:

1. Convert both numbers from n-base to decimal.
2. Add the two decimal numbers.
3. Convert the sum back to n-base.

Assumption:

$1 < n \leq 36$

Example

Input:

- 'n': '16'
- 'num1': "1A"
- 'num2': "B4"

Output:

- "CE"

Explanation:

- '1A' in decimal is '26', 'B4' in decimal is '180'. Their sum is '206', which in base 16 is "CE".
 -
-

Coding Question 15:

Digit Sum of a Number

Problem Statement:



The digit sum of a number is the sum of all its individual digits. For example, the digit sum of 1234 is $1 + 2 + 3 + 4 = 10$.

Implement the following function:

```
int DigitSum(int num);
```

The function accepts a positive integer num. Implement the function to calculate and return the sum of all its digits.

Steps:

1. Initialize a sum variable to zero.
2. Extract each digit from the number by repeatedly using the modulus operation ($\text{num} \% 10$) and add it to the sum.
3. Reduce the number by dividing it by 10 in each iteration ($\text{num} / 10$) until it becomes zero.
4. Return the sum of the digits.

Assumption:

- 'num' ≥ 0

Example

Input:

- 'num': '451'

Output:

- '10'

Explanation:

- $4 + 5 + 1 = 10$

Coding Question 16: Reverse Digits of a Number

Problem Statement:

Reversing the digits of a number means constructing a new number by reversing the order of its digits. For example, the reverse of 1234 is 4321.

Implement the following function:

```
int ReverseDigits(int num);
```

The function accepts a positive integer num. Implement the function to reverse the digits of the number and return the reversed number as an integer.



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Steps:

1. Initialize a variable to store the reversed number, starting from zero.
2. Extract each digit from the number by repeatedly using the modulus operation ($\text{num} \% 10$) and add it to the reversed number.
3. Shift the reversed number left by one position by multiplying it by 10, then add the extracted digit.
4. Reduce the number by dividing it by 10 in each iteration ($\text{num} / 10$) until it becomes zero.
5. Return the reversed number.

Assumption:

- 'num ≥ 0 '

Example

Input:

- 'num': '451'

Output:

- '154'

Explanation:

- The digits of 451 are reversed to form 154.

Coding Question 17: Count Vowels and Consonants

Problem Statement:

You are given a function:

```
void CountVowelsAndConsonants(const char str[], int* vowels, int* consonants);
```

The function accepts a string str and two integer pointers vowels and consonants. Implement the function to count the number of vowels and consonants in the given string and store the counts in the variables pointed to by vowels and consonants, respectively.

Steps:

1. Traverse the string from the beginning to the end.
2. For each character, check if it's a vowel ('a', 'e', 'i', 'o', 'u').
3. If it is a vowel, increment the vowels count.
4. If it's a consonant (a letter that isn't a vowel), increment the consonants count.

5. Ignore non-alphabetical characters.

Assumption:

- The string contains only lower-case alphabetical letters and possibly spaces or punctuation.
- The string is not null.

Example

Input:

- str: "apples and oranges"

Output:

- vowels: 6
- consonants: 10

Explanation:

- There are 6 vowels (a, e, a, o, a, e) and 10 consonants (p, p, l, s, n, d, r, n, g, s) in the string.

Coding Question 18: Rotate Characters in a String

Problem Statement:

You are given a function:

```
void RotateCharacters(char str[], int n, int k);
```

The function accepts a string str of length n and an integer k. Implement the function to rotate the characters of the string to the right by k positions. The rotation should be done in-place, modifying the original string.

Steps:

1. Calculate the effective rotation by taking $k \% n$ to handle cases where k is greater than the length of the string.
2. Reverse the entire string.
3. Reverse the first k characters.
4. Reverse the remaining characters from position k to the end of the string.

Assumption:

- The string contains only lower-case alphabetical letters.
- The string is not null.

Example



Input:

- str: "apples"
- k: 2

Output:

- "esappl"

Explanation:

- Rotating the string "apples" by 2 positions to the right gives "esappl".
-

Coding Question: 19

You are required to implement the following function, *Int OperationChoices(int c, int n, int a, int b)*

The function accepts 3 positive integers 'a', 'b' and 'c' as its arguments. Implement the function to return.

- (a + b), if c=1
- (a + b), if c=2
- (a * b), if c=3
- (a / b), if c=4

Assumption :

All operations will result in integer output.

Example:

- **Input**
 - c :1
 - a:12
 - b:16
- **Output:**
 - Since 'c'=1 , (12+16) is performed which is equal to 28 , hence 28 is returned.

Sample Input

c : 2

a : 16

b : 20

Sample Output

-4

Coding Question 20: Basic Arithmetic Operations

Problem Statement:

You are required to implement the following function:

```
int BasicArithmetic(int op, int x, int y);
```

The function accepts 3 integers op, x, and y. Implement the function to perform a specific arithmetic operation based on the value of op and return the result.

- Return $x + y$ if $op = 1$
- Return $x - y$ if $op = 2$
- Return $x * y$ if $op = 3$
- Return $x \% y$ if $op = 4$ (Assume $y \neq 0$)

Assumption:

All operations will result in integer output.

Example

Input:

- op: 2
- x: 20
- y: 5

Output:

- 15

Explanation:

- Since $op = 2$, $20 - 5$ is performed, resulting in 15.
-

Coding Question 21: Basic Calculator Operations

Problem Statement:

You are required to implement the following function:

```
int BasicCalculator(int choice, int x, int y);
```

The function accepts three integers choice, x, and y as its arguments. Implement the function to perform the operation based on the value of choice and return the result.

- Return $(x + y)$, if choice = 1
- Return $(x - y)$, if choice = 2
- Return $(x * y)$, if choice = 3
- Return (x / y) , if choice = 4
- Return $(x \% y)$, if choice = 5

Assumption:

All operations will result in integer output. Assume y is not zero for division and modulus operations.

Example:

Input:

- choice: 3
- x: 15
- y: 5

Output:

- 75

Explanation:

Since choice = 3, the multiplication $15 * 5$ is performed, resulting in 75.

Coding Question 22: Power or Modulus Operation

Problem Statement:

You are required to implement the following function:

`int PowerOrModulus(int choice, int base, int exponent);`

The function accepts three integers choice, base, and exponent as its arguments. Implement the function to perform the operation based on the value of choice and return the result.

- Return $\text{base}^{\text{exponent}}$ (power), if choice = 1
- Return $\text{base} \% \text{exponent}$ (modulus), if choice = 2

Assumption:

- base and exponent are positive integers.



- For the modulus operation, assume exponent is not zero.

Example:

Input:

- choice: 1
- base: 2
- exponent: 3

Output:

- 8

Explanation:

Since choice = 1, the power operation 2^3 is performed, resulting in 8.

Coding Question 23

You are required to implement the following function, *Int Calculate(int m, int n);*

The function accepts 2 positive integer 'm' and 'n' as its arguments. You are required to calculate the sum of numbers divisible both by 3 and 5, between 'm' and 'n' both inclusive and return the same.

Note:

$$0 < m \leq n$$

Example

Input:

m : 12

n : 50

Output:

90

Explanation:

The numbers divisible by both 3 and 5, between 12 and 50 both inclusive are {15, 30, 45} and their sum is 90.

Sample Input

m : 100

n : 160

Sample Output

405

Coding Question 24: Sum of Even Numbers in a Range

Problem Statement:

You are required to implement the following function:

```
int SumOfEvens(int m, int n);
```

The function accepts two positive integers m and n as its arguments. You are required to calculate the sum of all even numbers between m and n both inclusive and return the same.

Assumption:

- $0 < m \leq n < m \leq n$

Example:

Input:

- $m: 10$
- $n: 20$

Output:

- 90

Explanation:

The even numbers between 10 and 20 are {10, 12, 14, 16, 18, 20}, and their sum is 90.

Coding Question 25: Count Prime Numbers in a Range

Problem Statement:

You are required to implement the following function:

```
int CountPrimes(int m, int n);
```

The function accepts two positive integers m and n as its arguments. You are required to count the number of prime numbers between m and n both inclusive and return the count.

Assumption:

- $0 < m \leq n < m \leq n$

Example:

Input:

- $m: 10$

- n: 30

Output:

- 6

Explanation:

The prime numbers between 10 and 30 are {11, 13, 17, 19, 23, 29}, so the count is 6.

Coding Question 26

You are required to input the size of the matrix then the elements of matrix, then you have to divide the main matrix in two sub matrices (even and odd) in such a way that element at 0 index will be considered as even and element at 1st index will be considered as odd and so on. then you have sort the even and odd matrices in ascending order then print the sum of second largest number from both the matrices

Example:

- enter the size of array : 5
- enter element at 0 index : 3
- enter element at 1 index : 4
- enter element at 2 index : 1
- enter element at 3 index : 7
- enter element at 4 index : 9

Sorted even array : 1 3 9

Sorted odd array : 4 7 10

Coding Question 27

You are required to input the size of the matrix and then the elements of the matrix. Next, you have to divide the main matrix into two sub-matrices (even and odd) in such a way that the element at the 0th index will be considered even, and the element at the 1st index will be considered odd, and so on. After dividing, sort the even and odd matrices in descending order, then print the sum of the third largest number from both matrices.

Example:

- Enter the size of the array: 7
- Enter element at index 0: 12
- Enter element at index 1: 5
- Enter element at index 2: 8



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- Enter element at index 3: 3
- Enter element at index 4: 7
- Enter element at index 5: 15
- Enter element at index 6: 10

Output:

- Sorted even array: 12 10 8
- Sorted odd array: 15 7 5
- Third largest sum: $8 + 5 = 13$

Coding Question 28

You are required to input the size of the matrix and then the elements of the matrix. Next, you have to divide the main matrix into two sub-matrices (even and odd) in such a way that the element at the 0th index will be considered even, and the element at the 1st index will be considered odd, and so on. After dividing, sort the even and odd matrices in ascending order, then print the difference between the second smallest number from the even matrix and the second smallest number from the odd matrix.

Example:

- Enter the size of the array: 6
- Enter element at index 0: 9
- Enter element at index 1: 4
- Enter element at index 2: 6
- Enter element at index 3: 8
- Enter element at index 4: 2
- Enter element at index 5: 7

Output:

- Sorted even array: 2 6 9
- Sorted odd array: 4 7 8
- Difference: $6 - 7 = -1$

Coding Question : 29

Autobiographical Number



Problem Statement :

An Autobiographical Number is a number N such that the first digit of N represents the count of how many zeroes are there in N, the second digit represents the count of how many ones are there in N and so on.

You are given a function, def FindAutoCount(n):

The function accepts string "n" which is a number and checks whether the number is an autobiographical number or not. If it is, an integer is returned, i.e. the count of distinct numbers in 'n'. If not, it returns 0.

Assumption:

- The input string will not be longer than 10 characters.
- Input string will consist of numeric characters.

Note:

If string is None return 0.

Example:

Input:

n: "1210"

Output:

3

Explanation:

0th position in the input contains the number of 0 present in input, i.e. 1, in 1st position the count of number of 1s in input i.e. 2, in 2nd position the count of 2s in input i.e. 1, and in 3rd position the count of 3s i.e. 0, so the number is an autobiographical number.

Now unique numbers in the input are 0, 1, 2, so the count of unique numbers is 3. So 3 is returned.

Coding Question : 30

Self-Descriptive Number

Problem Statement:

A Self-Descriptive Number is a number N such that the digit in each position counts how many times a digit equal to that position appears in the number. For example, in the number 2020, the digit at index 0 (from the left) represents how many 0s are in the number, the digit at index 1 represents how many 1s, and so on.

You are given a function:

def FindSelfDescriptive(n: str) -> int:

The function accepts a string `n` which is a number and checks whether the number is self-descriptive or not. If it is, an integer is returned, representing the sum of the digits of `n`. If not, it returns 0.

Assumption:

- The input string will not be longer than 10 characters.
- The input string will consist of numeric characters.

Note:

- If the input string is None or empty, return 0.

Example:

Input:

- `n: "6210001000"`

Output:

- 10

Explanation:

- The 0th position has 6, meaning there are six 0s in the number, which is true.
- The 1st position has 2, meaning there are two 1s in the number, which is true.
- The 2nd position has 1, meaning there is one 2 in the number, which is true.
- The other digits follow similarly, confirming this is a self-descriptive number.

The sum of the digits in 6210001000 is $6 + 2 + 1 + 0 + 0 + 0 + 1 + 0 + 0 + 0 = 10$.

So, 10 is returned.

Coding Question : 31

Find the maximum value and its index in the array

Problem Statement :

You are given a function, `void MaxInArray(int arr[], int length)`; The function accepts an integer array 'arr' of size 'length' as its argument. Implement the function to find the maximum element of the array and print the maximum element and its index to the standard output

(STDOUT). The maximum element and its index should be printed in separate lines.

Note:

- Array index starts with 0



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- Maximum element and its index should be separated by a line in the output
- Assume there is only 1 maximum element in the array
- Print exactly what is asked, do not print any additional greeting messages

Example:

Input:

23 45 82 27 66 12 78 13 71 86

Output:

86

9

Explanation:

86 is the maximum element of the array at index 9.

Coding Question : 32

Instructions: You are required to write the code. You can click on compile & run anytime to check the compilation/ execution status of the program. The submitted code should be logically/syntactically correct and pass all the test cases.

Ques: The program is supposed to calculate the sum of the distance between three points from each other.

For

$x_1 = 1$ $y_1 = 1$

$x_2 = 2$ $y_2 = 4$

$x_3 = 3$ $y_3 = 6$

Distance is calculated as : $\text{sqrt}(x_2 - x_1)^2 + (y_2 - y_1)^2$

Coding Question : 33

Instructions: You are required to write the code. You can click on compile and run anytime to check compilation/execution status. The code should be logically/syntactically correct.

Question: Write a program in C such that it takes a lower limit and upper limit as inputs and print all the intermediate palindrome numbers.

Test Cases:

TestCase 1:

Input :

10 , 80

Expected Result:

11 , 22 , 33 , 44 , 55 , 66 , 77.



Test Case 2:

Input:

100,200

Expected Result:

101 , 111 , 121 , 131 , 141 , 151 , 161 , 171 , 181 , 191.

Coding Question : 34

Instructions: You are required to write the code. You can click on compile and run anytime to check the compilation/execution status. The code should be logically/syntactically correct.

Problem: Write a program in C to display the table of a number and print the sum of all the multiples in it.

Test Cases:

Test Case 1:

Input:

5

Expected Result Value:

5, 10, 15, 20, 25, 30, 35, 40, 45, 50

275

Test Case 2:

Input:

12

Expected Result Value:

12, 24, 36, 48, 60, 72, 84, 96, 108, 120

660

Coding Question : 35

Problem Statement

You are required to input the size of the matrix then the elements of matrix, then you have to divide the main matrix in two sub matrices (even and odd) in such a way that element at 0 index will be considered as even and element at 1st index will be considered as odd and so on. then you have sort the even and odd matrices in ascending order then print the sum of second largest number from both the matrices

Example

- enter the size of array : 5
- enter element at 0 index : 3
- enter element at 1 index : 4
- enter element at 2 index : 1



- enter element at 3 index : 7
- enter element at 4 index : 9

Sorted even array : 1 3 9

Sorted odd array : 4 7

7

Coding Question : 36

Problem Statement

You are required to implement the following function:

Int Calculate(int m, int n);

The function accepts 2 positive integer 'm' and 'n' as its arguments. You are required to calculate the sum of numbers divisible both by 3 and 5, between 'm' and 'n' both inclusive and return the same.

Note

$0 < m \leq n$

Example

Input:

m : 12

n : 50

Output

90

Explanation:

The numbers divisible by both 3 and 5, between 12 and 50 both inclusive are {15, 30, 45} and their sum is 90.

Sample Input

m : 100

n : 160

Sample Output

510

Coding Question : 37

Problem Statement

You are given a function,

Int MaxExponents (int a , int b);



You have to find and return the number between 'a' and 'b' (range inclusive on both ends) which has the maximum exponent of 2.

The algorithm to find the number with maximum exponent of 2 between the given range is

1. Loop between 'a' and 'b'. Let the looping variable be 'i'.
2. Find the exponent (power) of 2 for each 'i' and store the number with maximum exponent of 2 so far in a variable , let say 'max'. Set 'max' to 'i' only if 'i' has more exponent of 2 than 'max'.
3. Return 'max'.

Assumption: $a < b$

Note: If two or more numbers in the range have the same exponents of 2 , return the small number.

Example

- Input:
 - 7
 - 12
- Output:
 - 8

Explanation:

Exponents of 2 in:

7-0

8-3

9-0

10-1

11-0

12-2

Hence maximum exponent if two is of 8.

Coding Question : 38

Problem Statement

You are required to implement the following function.

Int OperationChoices(int c, int n, int a , int b)

The function accepts 3 positive integers 'a', 'b' and 'c' as its arguments. Implement the function to return.

- $(a + b)$, if $c=1$
- $(a - b)$, if $c=2$
- $(a * b)$, if $c=3$
- (a / b) , if $c=4$

Assumption : All operations will result in integer output.

Example:

- Input
 - $c:1$
 - $a:12$
 - $b:16$
- Output:
 - Since ' $c=1$ ', $(12+16)$ is performed which is equal to 28, hence 28 is returned.

Sample Input

$c:2$

$a:16$

$b:20$

Sample Output

-4

Coding Question : 39

Problem Statement

You are given a function,

`Void *ReplaceCharacter(Char str[], int n, char ch1, char ch2);`

The function accepts a string 'str' of length n and two characters 'ch1' and 'ch2' as its arguments. Implement the function to modify and return the string 'str' in such a way that all occurrences of 'ch1' in original string are replaced by 'ch2' and all occurrences of 'ch2' in original string are replaced by 'ch1'.

Assumption: String Contains only lower-case alphabetical letters.

Note:



- Return null if string is null.
- If both characters are not present in string or both of them are same , then return the string unchanged.

Example:

- Input:
 - Str: apples
 - ch1:a
 - ch2:p
- Output:
 - paaales

Explanation:

'A' in original string is replaced with 'p' and 'p' in original string is replaced with 'a', thus output is paaales.

Coding Question : 40

Problem Statement

A carry is a digit that is transferred to left if sum of digits exceeds 9 while adding two numbers from right-to-left one digit at a time

You are required to implement the following function.

Int NumberOfCarries(int num1 , int num2);

The functions accepts two numbers 'num1' and 'num2' as its arguments. You are required to calculate and return the total number of carries generated while adding digits of two numbers 'num1' and ' num2'.

Assumption: num1, num2>=0

Example:

- Input
 - Num 1: 451
 - Num 2: 349
- Output
 - 2

Explanation:



Adding 'num 1' and 'num 2' right-to-left results in 2 carries since (1+9) is 10. 1 is carried and (5+4=1) is 10, again 1 is carried. Hence 2 is returned.

Sample Input

Num 1: 23

Num 2: 563

Sample Output

0

Coding Question : 41

Implement the following functions.a

`char*MoveHyphen(char str[],int n);`

The function accepts a string "str" of length 'n', that contains alphabets and hyphens (-). Implement the function to move all hyphens(-) in the string to the front of the given string.

NOTE:- Return null if str is null.

Example :-

- Input:
 - str.Move-Hyphens-to-Front
- Output:
 - —MoveHyphenstoFront

Explanation:-

The string "Move-Hyphens -to-front" has 3 hyphens (-), which are moved to the front of the string, this output is "— MoveHyphen"

Sample Input

- Str: String-Compare

Sample Output-

- -StringCompare

Coding Question : 42

N-base notation is a system for writing numbers that uses only n different symbols, This symbols are the first n symbols from the given notation list(Including the symbol for o) Decimal to n base notation are (0:0, 1:1, 2:2, 3:3, 4:4, 5:5, 6:6, 7:7, 8:8, 9:9, 10:A,11:B and so on upto 35:Z)



Implement the following function

Char* DectoNBase(int n, int num):

The function accept positive integer n and num Implement the function to calculate the n-base equivalent of num and return the same as a string

Steps:

1. Divide the decimal number by n,Treat the division as the integer division
2. Write the the remainder (in n-base notation)
3. Divide the quotient again by n, Treat the division as integer division
4. Repeat step 2 and 3 until the quotient is 0
5. The n-base value is the sequence of the remainders from last to first

Assumption:

$1 < n \leq 36$

Example

Input

n: 12

num: 718

Output

4BA

Explanation

num	Divisor	quotient	remainder
718	12	59	10(A)
59	12	4	11(B)
4	12	0	4(4)

Sample Input

n: 21

num: 5678

Sample Output

CI8

Coding Question : 43

Implement the following Function

def ProductSmallestPair(sum, arr)



The function accepts an integers sum and an integer array arr of size n. Implement the function to find the pair, (arr[j], arr[k]) where $j \neq k$, Such that arr[j] and arr[k] are the least two elements of array (arr[j] + arr[k] $\leq$ sum) and return the product of element of this pair

NOTE

- Return -1 if array is empty or if $n < 2$
- Return 0, if no such pairs found
- All computed values lie within integer range

Example

Input

sum:9

size of Arr = 7

Arr:5 2 4 3 9 7 1

Output

2

Explanation

Pair of least two element is (2, 1) $2 + 1 = 3 < 9$, Product of (2, 1) $2 * 1 = 2$. Thus, output is 2

Sample Input

sum:4

size of Arr = 6

Arr:9 8 3 -7 3 9

Sample Output

-21

Coding Question : 44

You are required to implement the following Function

def LargeSmallSum(arr)

The function accepts an integers arr of size 'length' as its arguments you are required to return the sum of second largest element from the even positions and second smallest from the odd position of given 'arr'

Assumption:

- All array elements are unique



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- Treat the 0th position as even

NOTE

- Return 0 if array is empty
- Return 0, if array length is 3 or less than 3

Example

Input

arr:3 2 1 7 5 4

Output

7

Explanation

- Second largest among even position elements(1 3 5) is 3
- Second smallest among odd position element is 4
- Thus output is $3+4 = 7$

Sample Input

arr:1 8 0 2 3 5 6

Sample Output

8

Coding Question : 45

Implement the following Function

def differenceofSum(n, m)

The function accepts two integers n, m as arguments Find the sum of all numbers in range from 1 to m(both inclusive) that are not divisible by n. Return difference between sum of integers not divisible by n with sum of numbers divisible by n.

Assumption:

- $n > 0$ and $m > 0$
- Sum lies between integral range

Example

Input

n:4

m:20



Output

90

Explanation

- Sum of numbers divisible by 4 are $4 + 8 + 12 + 16 + 20 = 60$
- Sum of numbers not divisible by 4 are $1 + 2 + 3 + 5 + 6 + 7 + 9 + 10 + 11 + 13 + 14 + 15 + 17 + 18 + 19 = 150$
- Difference $150 - 60 = 90$

Sample Input

n:3

m:10

Sample Output

19

Coding Question : 46

You are given a function,

`int findCount(int arr[], int length, int num, int diff);`

The function accepts an integer array 'arr', its length and two integer variables 'num' and 'diff'. Implement this function to find and return the number of elements of 'arr' having an absolute difference of less than or equal to 'diff' with 'num'.

Note: In case there is no element in 'arr' whose absolute difference with 'num' is less than or equal to 'diff', return -1.

Example:

Input:

- arr: 12 3 14 56 77 13
- num: 13
- diff: 2

Output:

3

Explanation:

Elements of 'arr' having absolute difference of less than or equal to 'diff' i.e. 2 with 'num' i.e. 13 are 12, 13 and 14.

Coding Question : 47

You are given a function.

`int CheckPassword(char str[], int n);`

The function accepts string str of size n as an argument. Implement the function which returns 1 if

given string str is valid password else 0.
str is a valid password if it satisfies the below conditions.

- – At least 4 characters
- – At least one numeric digit
- – At Least one Capital Letter
- – Must not have space or slash (/)
- – Starting character must not be a number

Assumption:

Input string will not be empty.

Example:

Input 1:

aA1_67

Input 2:

a987 abC012

Output 1:

1

Output 2:

0

Coding Question : 48

Problem Description :

The Binary number system only uses two digits, 0 and 1 and number system can be called binary string. You are required to implement the following function:

`int OperationsBinaryString(char* str);`

The function accepts a string str as its argument. The string str consists of binary digits eparated with an alphabet as follows:

- – A denotes AND operation
- – B denotes OR operation
- – C denotes XOR Operation

You are required to calculate the result of the string str, scanning the string to right taking one operation at a time, and return the same.

Note:

- No order of priorities of operations is required
- Length of str is odd



- If str is NULL or None (in case of Python), return -1

Input:

str: 1C0C1C1A0B1

Output:

1

Explanation:

The alphabets in str when expanded becomes "1 XOR 0 XOR 1 XOR 1 AND 0 OR 1", result of the expression becomes 1, hence 1 is returned.

Sample Input:

0C1A1B1C1C1B0A0

Output:

0

Coding Question : 49

Problem Description :

The function accepts two positive integers 'r' and 'unit' and a positive integer array 'arr' of size 'n' as its argument 'r' represents the number of rats present in an area, 'unit' is the amount of food each rat consumes and each ith element of array 'arr' represents the amount of food present in 'i+1' house number, where $0 \leq i$

Note:

- Return -1 if the array is null
- Return 0 if the total amount of food from all houses is not sufficient for all the rats.
- Computed values lie within the integer range.

Example:

Input:

- r: 7
- unit: 2
- n: 8
- arr: 2 8 3 5 7 4 1 2

Output:

4

Explanation:

Total amount of food required for all rats = $r * \text{unit}$



$$= 7 * 2 = 14.$$

The amount of food in 1st houses = $2+8+3+5 = 18$. Since, amount of food in 1st 4 houses is sufficient for all the rats. Thus, output is 4.

Coding Question 50

Instructions: You are required to write the code. You can click on compile and run anytime to check the compilation/execution status. The code should be logically/syntactically correct.

Question: Write a program in C that takes a lower limit and an upper limit as inputs and prints all the intermediate prime numbers within the given range.

Test Cases:

Test Case 1:

- Input:
10, 30
- Expected Result:
11, 13, 17, 19, 23, 29

Test Case 2:

- Input:
50, 100
- Expected Result:
53, 59, 61, 67, 71, 73, 79, 83, 89, 97